# Performing CRUD operations in DynamoDB using AWS Lambda

In this assignment I have created a lambda function in AWS which gets triggered by the API gateway to perform the CRUD operations (POST, GET, PUT, DELETE) in DynamoDB.

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AWS Services used:

1. *Lambda*: AWS Lambda is a serverless function which is event-driven and does not need pre-provisioned servers to execute the code. It gets triggered in response to certain events while automatically managing the computing resources.

2. *DynamoDB*: DynamoDB is a serverless NoSQL database which is highly scalable and can handle a varied amount of traffic. It is a key-value datastore with built-in support for replication.

3. *API Gateway*: API Gateway is a service which is used to build and monitor scalable APIs to connect the applications to various backend services.

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Lambda function Code:

lambda_function.py:

```
import json
import boto3
from boto3.dynamodb.conditions import Key

dynamodb = boto3.resource('dynamodb')
student_table = dynamodb.Table('StudentRecords')

def lambda_handler(event, context):

    if event['httpMethod'] == 'POST':
        # create student record
        student_record = json.loads(event['body'])
        student_table.put_item(Item = student_record)

        return {
            'statusCode': 200,
            'body': json.dumps('Student record added successfully')
        }

    elif event['httpMethod'] == 'GET':
        # fetch student record by student id
```

```
    student_id = event['queryStringParameters']['student_id']
    response = student_table.get_item(Key={'student_id': student_id})

    if 'Item' not in response:
        # if record is not present for the given student id
        return {
            'statusCode': 404,
            'body': json.dumps(f'Error: Record not found for student id: {student_id}')
        }

    return {
        'statusCode': 200,
        'body': json.dumps(response['Item'])
    }

elif event['httpMethod'] == 'PUT':
    # update student record by student id
    student_record = json.loads(event['body'])

    student_table.update_item(
        Key={'student_id': student_record['student_id']},
        UpdateExpression="set #name=:n, #course=:c",
        ExpressionAttributeNames={
            '#name': 'name',
            '#course': 'course'
        },
        ExpressionAttributeValues={
            ':n': student_record['name'],
            ':c': student_record['course']
        }
    )

    return {
        'statusCode':200,
        'body': json.dumps('Student record updated successfully')
    }

elif event['httpMethod'] == 'DELETE':
    # delete student record
    student_id = event['queryStringParameters']['student_id']

    response = student_table.get_item(Key={'student_id': student_id})
```

```
if 'Item' not in response:
    # if record is not present for the given student id
    return {
        'statusCode': 404,
        'body': json.dumps(f'Error: Record not found for student id: {student_id}')
    }

student_table.delete_item(Key={'student_id': student_id})

return {
    'statusCode':200,
    'body': json.dumps('Student record deleted successfully')
}
```

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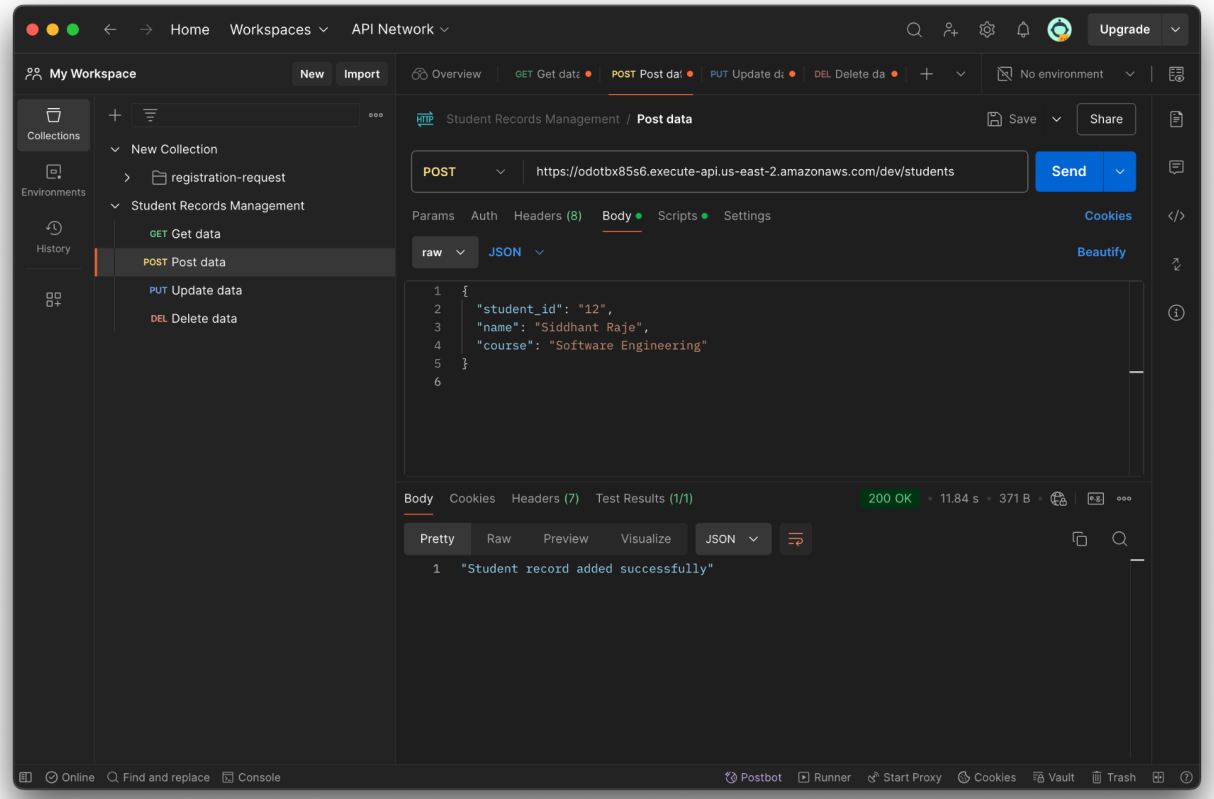
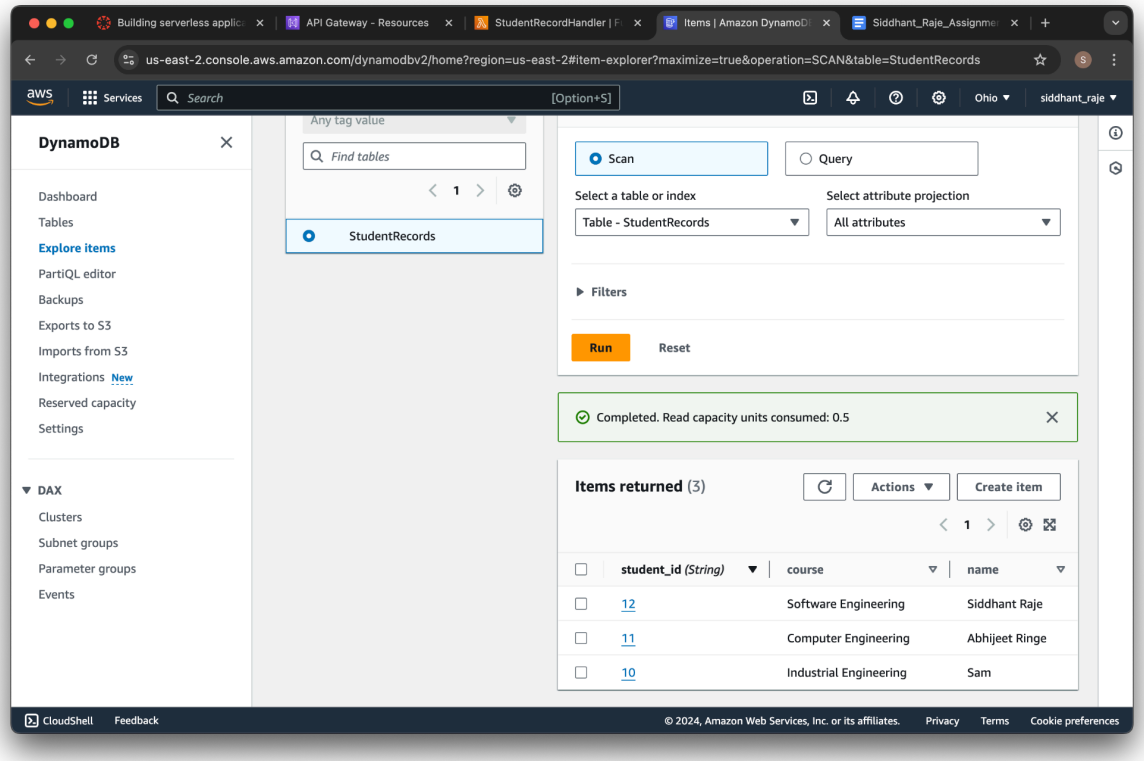
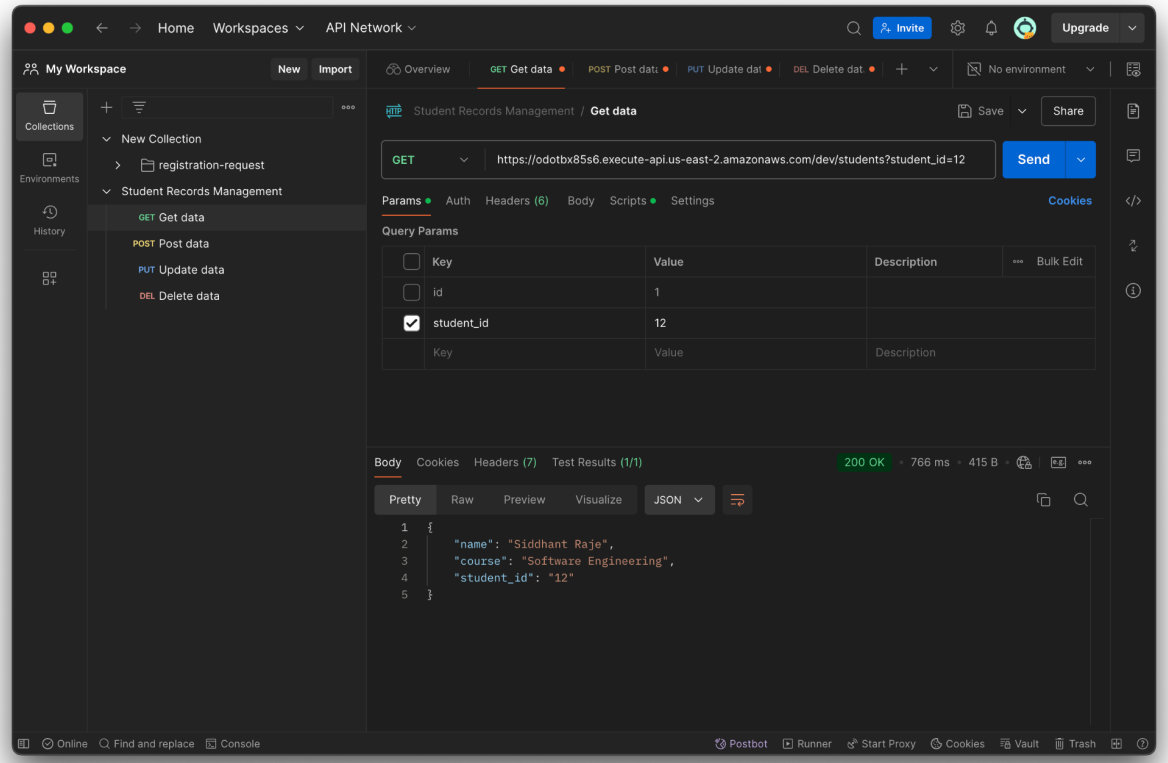
Screenshots:

1. Adding records (POST):

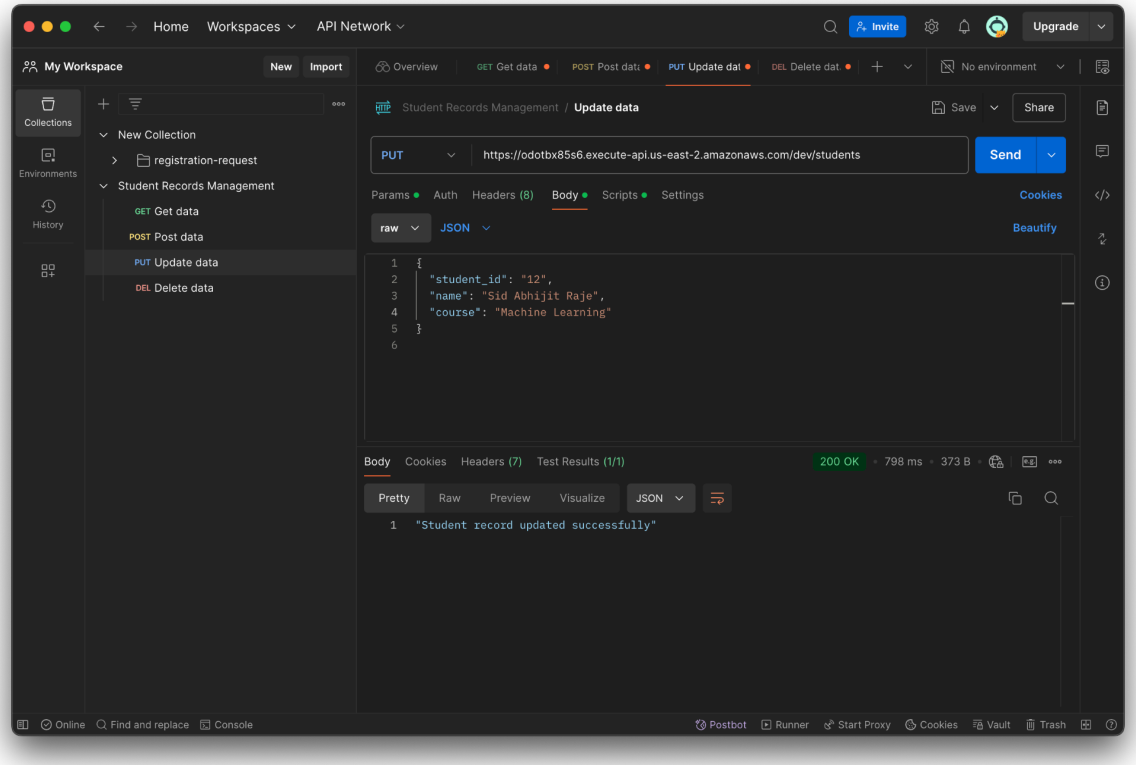


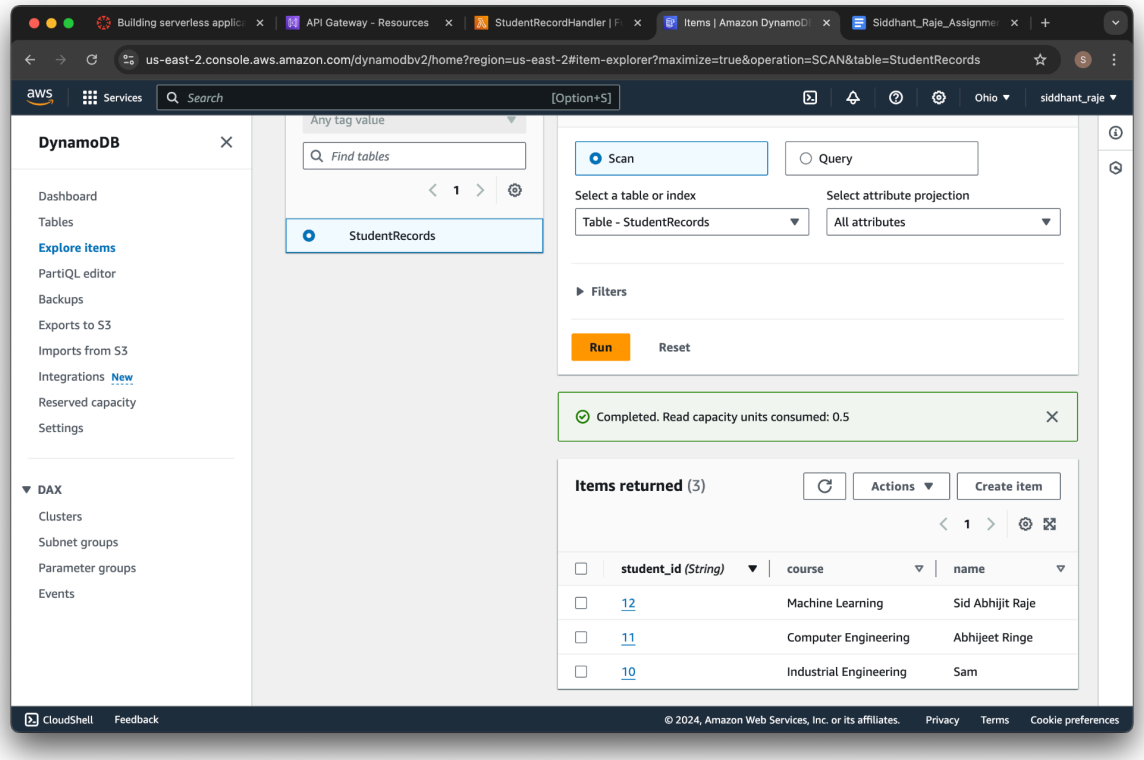



2. Fetching record by student_id (GET):

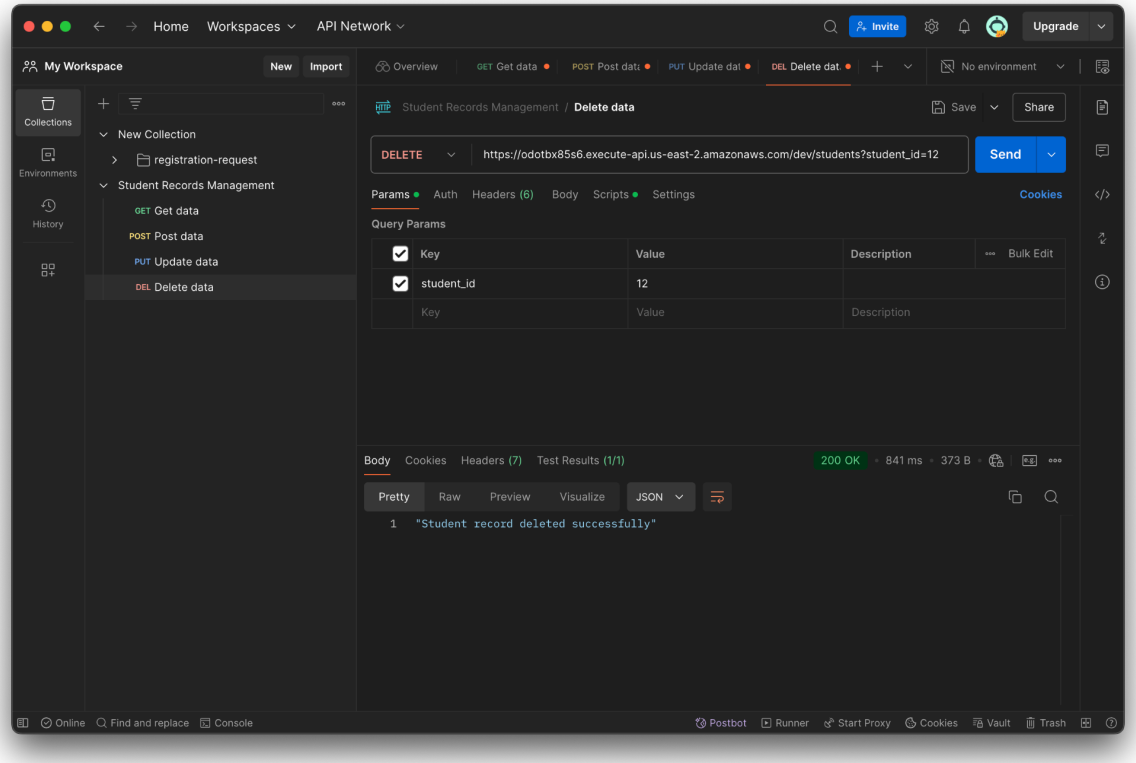


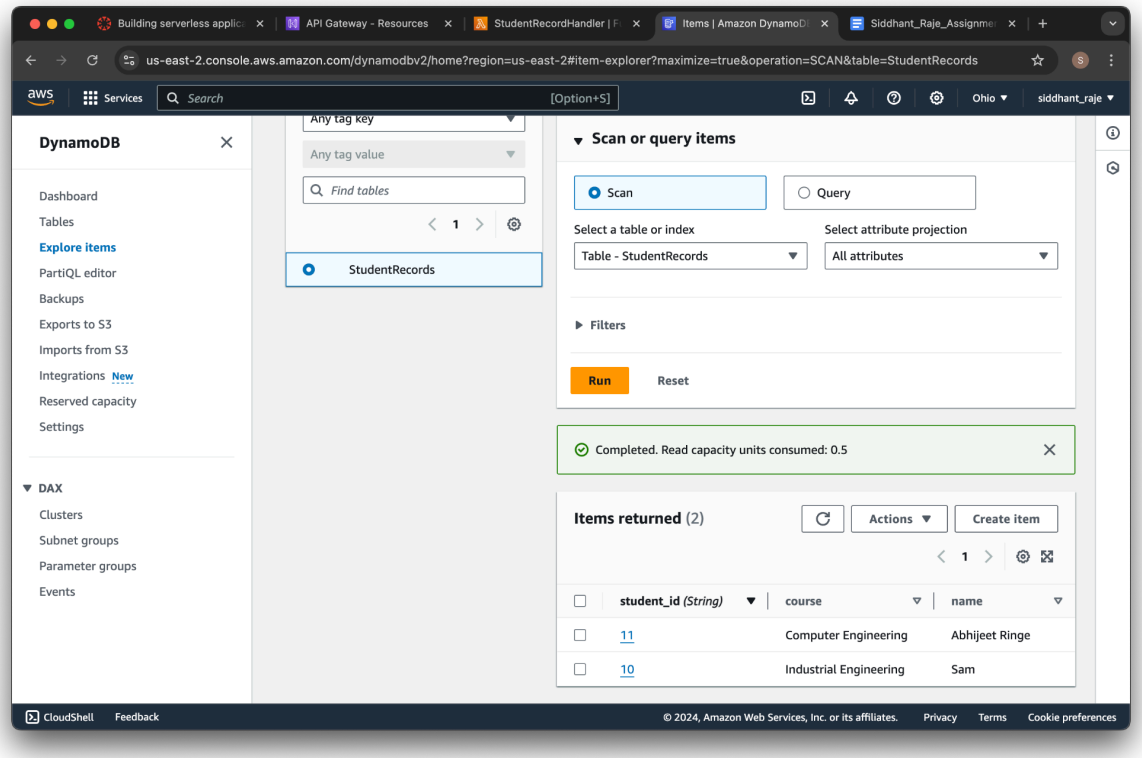
3. Updating the data (PUT):



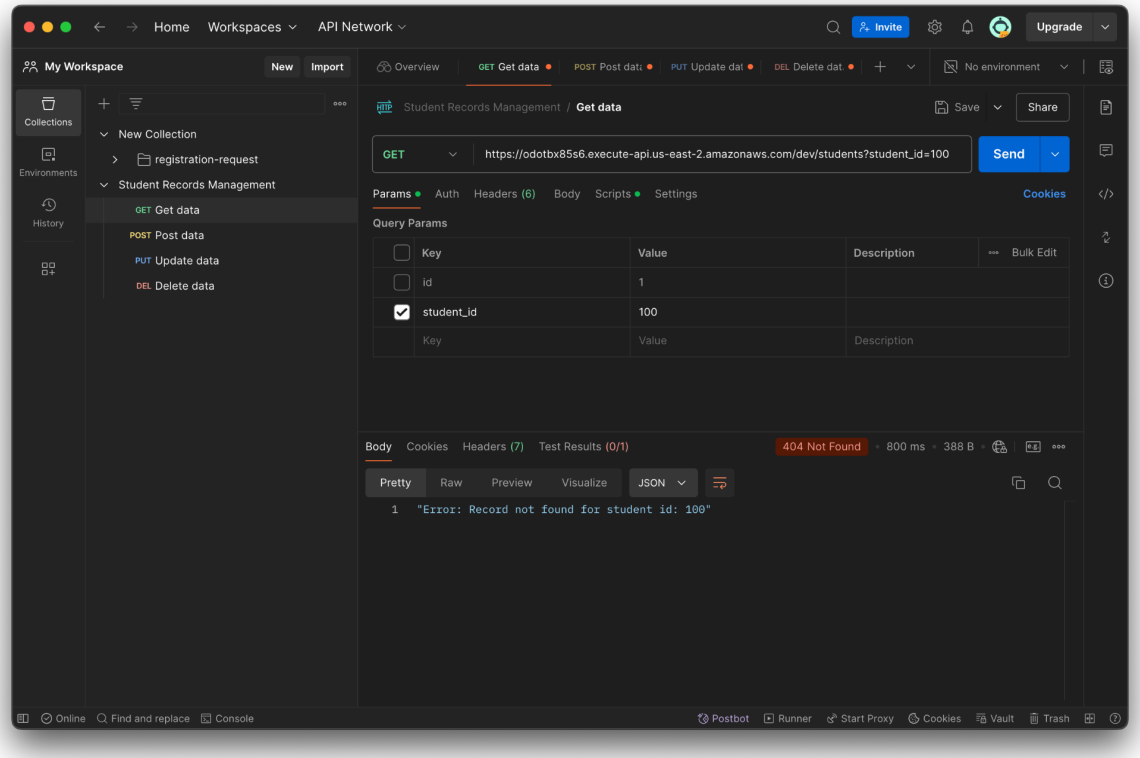



4. Deleting a record (DELETE):

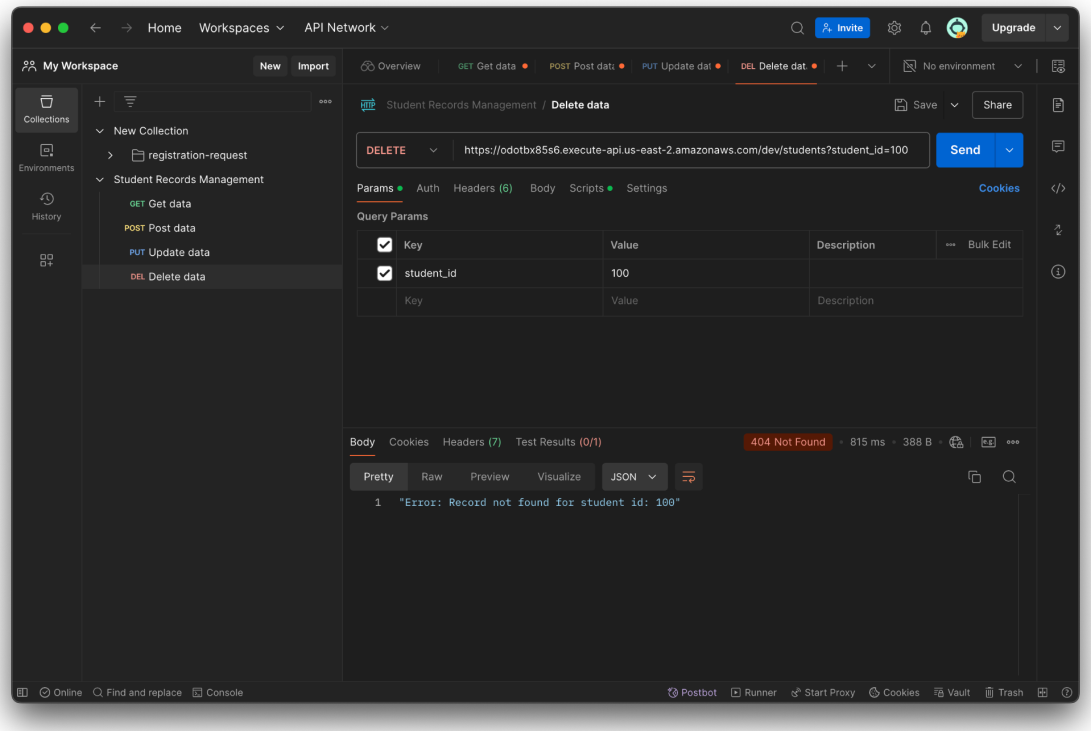


5. Edge Case - Trying to **fetch** a record which **does not exist**:




6. Edge Case - Trying to **delete** a record which **does not exist**:



Challenges and Learnings:

1. Challenges:

a. *Problem*: After building the lambda function and creating respective APIs for CRUD operations, when I tried to test the data in POSTMAN, I was getting a 'KeyError' - 'httpMethod'

*Observation*: After going through and debugging the code, I found out that the 'event' object was empty.

*Solution*: While creating the APIs, I turned on the 'Lambda Proxy Integration' in the 'Integration request' option which ensured that the API gateway would send the requests as a structured event.

*b. Problem*: While testing various CRUD operations, when a student_id which is non-existent is being given in the GET or DELETE request, an empty response was getting in return.

*Observation*: Edge cases for GET and DELETE operations were not handled in the Lambda function.

*Solution*: Added the code to first check if the given student_id exists in the database or not. If it does not exist, it should return a response which addresses the situation to the user.

2. Learnings:

a. Firstly I learnt about the serverless architecture, its pros and cons, and the use cases of a serverless architecture. I understood the applications and the best practices of using this architecture.
b. I learnt how to set up and configure the AWS Lambda to trigger based on REST API calls and perform CRUD operations on DynamoDB.
c. I researched about DynamoDB and its use cases and understood when a NoSQL database should be used. I got hands-on experience on performing various operations in the DynamoDB.
d. I gained insights on creating REST APIs using the AWS API Gateway.
e. I also understood the importance of cloud computing and various features like Scalability, Availability, Security etc. which is an integral part of any Enterprise level software.